



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

AGRICULTURAL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 11 Agricultural Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing units. Accept other correct answers.

QUESTION 1

- 1.1 1.1.1 A ✓✓ 1.1.2 A ✓✓ 1.1.3 C ✓✓ 1.1.4 B ✓✓ 1.1.5 A ✓✓ (10)
- 1.2 1.2.1 Hydraulics (hydraulic system) ✓✓ 1.2.2 Head (lift height) ✓✓ 1.2.3 (10)
Thermoplastic ✓✓ 1.2.4 Preventive (scheduled) maintenance ✓✓ 1.2.5 Transformer
✓✓

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QUESTION 2

- 2.1 Thermoplastics soften when heated and can be reshaped many times ✓, e.g. (4)
polythene, PVC, nylon ✓. Thermosetting plastics set permanently when first heated and
cannot be remelted ✓, e.g. Bakelite, epoxy resin, melamine ✓.
- 2.2 Any THREE: irrigation pipes ✓; tunnel / greenhouse covers ✓; water tanks ✓; silage (3)
wrap; mulch sheeting; feed bags; ear tags
- 2.3 MIG, any ONE: fast, continuous welding; little slag to clean; easy to learn; good for thin (4)
metal ✓✓. Arc, any ONE: cheap and portable equipment; works outdoors in wind;
suitable for thick and dirty metal ✓✓.
- 2.4 Any TWO with a cause: porosity (gas holes) ✓ → dirty metal or damp electrodes ✓; (4)
undercut ✓ → current too high or wrong angle ✓; slag inclusion → slag not cleaned
between runs; cracks → cooling too fast; lack of fusion → current too low or welding too
fast
- 2.5 Cast iron is brittle ✓. Preheating reduces the temperature difference between the weld (3)
and the rest of the metal ✓, so it cools slowly and evenly and does not crack ✓.
- 2.6 Any TWO: visual inspection ✓; dye penetrant test ✓; X-ray (radiographic) test; (2)
ultrasonic test; magnetic particle test

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QUESTION 3

- 3.1 Any FOUR: engine oil level ✓; coolant (radiator) level ✓; tyre pressure and condition ✓; (4)
fuel level ✓; hydraulic oil level; air filter; brakes and lights; leaks
- 3.2 Pressure applied to a liquid in a closed system is passed on equally in all directions ✓✓. (4)
The hydraulic pump pushes oil into a cylinder; the pressure acts on a large piston and
creates a large force ✓ that lifts the three-point linkage and the implement ✓.
- 3.3 Any THREE: keep the PTO shaft guards in place ✓; switch off the PTO and engine (3)
before working on the implement ✓; no loose clothing near the shaft ✓; never step over
a turning shaft; keep bystanders away

- 3.4 (a) $45 \div 6 \checkmark = 7,5 \text{ days } \checkmark$ (5)
 (b) $45 \times 12 = 540 \text{ } \ell \checkmark$; $540 \times R23 \checkmark = R12\,420 \checkmark$
- 3.5 Correct pressure gives the best grip (traction) and less soil compaction $\checkmark$, reduces tyre wear and fuel use $\checkmark$. (2)
- 3.6 Any TWO: mouldboard plough $\checkmark$; disc plough $\checkmark$; chisel plough; subsoiler; ripper (2)

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QUESTION 4

- 4.1 Any FOUR: energiser $\checkmark$; earth system (earth spikes) $\checkmark$; insulators $\checkmark$; fence wires $\checkmark$; lightning diverter; warning signs; cut-out switches (4)
- 4.2 The circuit is completed through the soil and the earth system $\checkmark$; poor earthing gives a weak shock, so animals are not kept in $\checkmark$. (2)
- 4.3 (a) $P = VI = 230 \times 8,7 \checkmark = 2\,001 \text{ W } (\approx 2 \text{ kW}) \checkmark$ (6)
 (b) $2,001 \text{ kW} \times 5 \text{ h } \checkmark = 10,01 \text{ kWh } \checkmark$
 (c) $10,01 \times R2,80 \checkmark = R28,03 \checkmark$
- 4.4 Animal manure (and plant waste) is mixed with water in a sealed tank $\checkmark$. Bacteria break it down without oxygen (anaerobic digestion) $\checkmark$, producing gas that collects at the top $\checkmark$. Products: biogas (methane) for cooking, heating or electricity $\checkmark$; slurry used as fertiliser $\checkmark$. (5)
- 4.5 Any TWO: free energy from the sun $\checkmark$; works far from the electricity grid and during load shedding $\checkmark$; low maintenance; no fuel costs; no pollution (2)
- 4.6 It cuts off the power quickly when current leaks to earth, protecting people from electric shock $\checkmark$. (1)

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QUESTION 5

- 5.1 $4 + 26 \checkmark = 30 \text{ m } \checkmark$ (2)
- 5.2 $p = 1\,000 \times 9,8 \times 30 \checkmark\checkmark = 294\,000 \text{ Pa} = 294 \text{ kPa } \checkmark$ (3)
- 5.3 Centre pivot, any ONE: irrigates very large areas automatically with little labour $\checkmark\checkmark$. Drip, any ONE: uses the least water / water goes straight to the roots / fewer weeds $\checkmark\checkmark$. (4)
- 5.4 Too much water makes weak, porous concrete that cracks $\checkmark\checkmark$; too little water makes concrete hard to work and it does not set properly $\checkmark$. (3)
- 5.5 Concrete is weak in tension $\checkmark$; the steel mesh strengthens it and prevents cracking when the trough is full of water $\checkmark$. (2)
- 5.6 Any THREE: a large catchment area with enough runoff $\checkmark$; a narrow valley for a short, cheap wall $\checkmark$; soil that holds water (clay) $\checkmark$; solid foundation for the wall; close to where the water is needed; does not flood important land (3)
- 5.7 A centrifugal pump creates low pressure to suck water up $\checkmark$. If air leaks in, the pump loses its prime $\checkmark$ and cannot lift water (or it delivers less and can be damaged) $\checkmark$. (3)

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TOTAL: 100

END OF MARKING GUIDELINE